

HVIA Data Analysis Training Task

Olist Business Discovery | Sections A + B + C + E | Preview

Reading path: A Company -> B Dataset -> C Analysis -> E HVIA Solutions. Still to come: Outreach (F), LinkedIn (H).

A. Company Research

Olist helps Brazilian SMB retailers sell more through technology. Below: who they are, how money worked in the dataset era, what was added later, then a dated timeline. The public order extract (Kaggle ~2018) reflects the marketplace aggregator / Olist Store model - not today's full SaaS + fintech stack. Keep that split while reading B and C.

A.1 Snapshot

Field	Detail
Legal name	Olist Servicos Digitais Ltda. (CNPJ 18.552.346/0001-68)
Founded	2014 (commercially active from 2015)
Founder & CEO	Tiago Dalvi
HQ	Curitiba, Parana, Brazil (+ Sao Paulo, Bento Goncalves)
Team size	~790-850 (after SaaS-oriented restructuring)
Valuation / funding	\$1.5B unicorn (Dec 2021) \$320M+ (Goldman, SoftBank, Wellington, etc.)
Merchants / scale	50,000+ merchants ~R\$60-70B ecosystem invoicing (company-reported)
Primary customer	SMB retailers / brands (sellers); end consumers are demand-side in the CSVs

A.2 Dataset-era model (2016-2018) - what the data describes

Flow: small seller joins Olist -> products listed under Olist's verified storefront on major marketplaces -> customer buys from "Olist" -> seller fulfills. Monetization: roughly 19-21% blended commission (marketplace + Olist + payment-related costs) plus modest seller subscription / membership fees.

- Thin intermediary margins: historically gross margins in a mid-30s percent range were hard to expand while operating the aggregator layer.
- Reputation contagion: a few late or poor sellers could damage the shared Olist account rating on Mercado Livre, Amazon, etc., hurting thousands of good sellers at once.
- Merchant churn at scale: as seller GMV grew, a ~20% take-rate pushed sellers to open direct marketplace accounts; platforms also made direct onboarding easier over time.

A.3 Later model (context only) - what was added after the dataset

Olist evolved into a retail technology ecosystem often framed as a "Commerce OS": Tiny ERP, Vnda (D2C), Olist Pax (logistics), Flip (credit), PDV, and Lis (embedded AI). Indicative later mix: ~65% recurring SaaS, ~20% fintech/credit, ~10% logistics spread. Use this for company research; do not force 2025-2026 products into every 2016-2018 KPI.

- Competitors (examples): Bling, Nuvemshop, VTEX, Melhor Envio, Magazord - different ERP / storefront / shipping mixes.
- Stack (high level): Python/Go/React; Kafka; PostgreSQL/Redis/Elasticsearch; AWS + Spark/Databricks-style analytics.

A.4 Sequential timeline - what was added when

Read top to bottom. The 2016-18 band is locked to the public CSV extract. Later rows explain strategic evolution; they are not additional columns in the order tables.

2007	Solidarium: physical retail network supporting artisans and small producers (founder Tiago Dalvi's pre-Olist chapter - same SMB DNA).
2014-15	Olist founded. Insight: SMBs struggle to reach customers on major marketplaces, not only to open a store. Olist launches as a tech + logistics bridge / aggregator.
2016-18	DATASET WINDOW (this task). Olist Store / shared storefront: sellers fulfill under one Olist face; commission ~19-21% + fees; ~99k public orders. Shared reputation is the structural risk. All Sections B-C interpret operations in this window.
2019-20	Ecosystem build: Clickspace (marketplace / social commerce infrastructure) and PAX Logtech (later Olist Pax - carrier aggregation, rate shopping, discounted labels).
2021	Adds Tiny ERP (catalog, inventory, Brazilian tax invoicing) and Vnda (D2C / headless storefront). Unicorn at \$1.5B (Dec 2021). Software becomes a clearer growth path.
2022-24	Tech-winter discipline: cost control, focus on higher-margin SaaS recurring revenue, and operating leverage versus pure marketplace take-rate economics.
Aug 2025	Adds fintech Flip -> Flip by Olist for merchant working capital, supported by an FIDC credit structure (~R\$90M). Underwriting edge: ERP sales/invoice signals.
2026	Aug: reported operational breakeven path. Sep: CEO announces wind-down of the shared marketplace storefront model and a full shift toward SaaS + fintech - closing the chapter that the 2016-2018 dataset documents.

A.5 Takeaway - how to use this in B and C

Acknowledge today's Commerce OS in company research. For dataset understanding and analysis, stay inside 2016-2018: shared reputation, commission on orders, seller-fulfilled shipping, freight and ETA as customer-visible promises. Early dataset signals already point to late-delivery review collapse, near-zero repeat purchase, and seller revenue concentration - expanded next.

Bridge KPI (from CSVs)	Signal
~99.4k orders / ~96k unique	Scale of the public extract
Late vs ETA 6.8% but reviews 4.29 -> 2.27	Reputation risk > volume risk
96.9% one-time buyers	Acquisition-led growth, weak LTV
Top 10% sellers -> ~67.5% product revenue	GMV + quality concentration

B. Dataset Understanding

Public Brazilian E-Commerce extract (Olist Store). Purchase timestamps span roughly 2016-09-04 to 2018-10-17. Nine related CSVs. Not in scope: marketplace ad spend, seller subscription invoices, or post-2018 SaaS / Flip product telemetry.

B.1 Nine tables (structure)

Table	Rows / role
orders	99,441 Hub: status + purchase/approval/carrier/delivery/ETA
order_items	112,650 Line items: product, seller, price, freight (~1.14/order)
customers	99,441 Location; customer_unique_id = true person (96,096)
sellers	3,095 Merchant location
products	32,951 Category, photos, weight/dims
payments	103,886 Type, installments, value (some split payments)
reviews	99,224 Score 1-5 + optional text (~1 per order)
geolocation	1,000,163 Lat/lng; many rows per ZIP -> mean before distance
category_translation	71 Portuguese category -> English

B.2 How to join (sequential)

Step 1 | Start at orders

Filter/status first. ~97% delivered. Delivery KPIs need delivered + non-null delivered_customer_date (96,470 orders in the SLA cut).

Step 2 | Attach items -> products -> sellers

Revenue and freight live on items; aggregate to order before AOV or late-rate. Category via products + translation (610 products lack category; 13 untranslated).

Step 3 | Attach customers + payments

Retention only via customer_unique_id. Payments: ~74% credit card, ~19% boleto, ~6% voucher; ~3k orders have multiple payment rows.

Step 4 | Attach reviews + geo

Aggregate review_score if duplicate order_id (551 duplicate rows). ZIP geo is not unique - mean lat/lng per prefix before seller-customer distance.

B.3 Data quality + exploration snapshot

Main traps are grain mismatches (item vs order) and geo duplication - not corrupt CSVs. Review titles missing ~88% and messages ~59%: scores scale; text themes are biased toward customers who wrote comments.

KPI	Value
Product revenue / freight / AOV	R\$13.6M R\$2.3M ~R\$138
Late vs ETA (delivered)	6.8% of 96,470
Review on-time vs late	4.29 vs 2.27 stars
One-time buyers	96.9%
Top 10% sellers -> revenue	~67.5%
SP share (customers / sellers)	~42% / ~60%
Cancel + unavailable	~1.24% (secondary vs late & loyalty)

Takeaway: order-centric marketplace ops data. Fulfillment reliability and seller heterogeneity sit on a one-shot customer base - the cause chain in Section C.

C. Analysis & Business Story | Cause chain

Read as a sequence: growth stress -> logistics and geography -> late deliveries -> review collapse and written complaints -> weak repeat purchase - amplified by a concentrated seller base under one shared Olist face.

C.1 Growth created operational pressure

Orders scaled fast (Jan 2017 -> Jan 2018 YoY ~+809%, with an early-base caveat: Jan 2017 had only ~800 orders). Peak month Nov 2017: 7,544 orders (Black Friday context). Cancel + unavailable together ~1.2% - secondary versus lateness and loyalty. Peaks expose SLA gaps faster than quiet days.

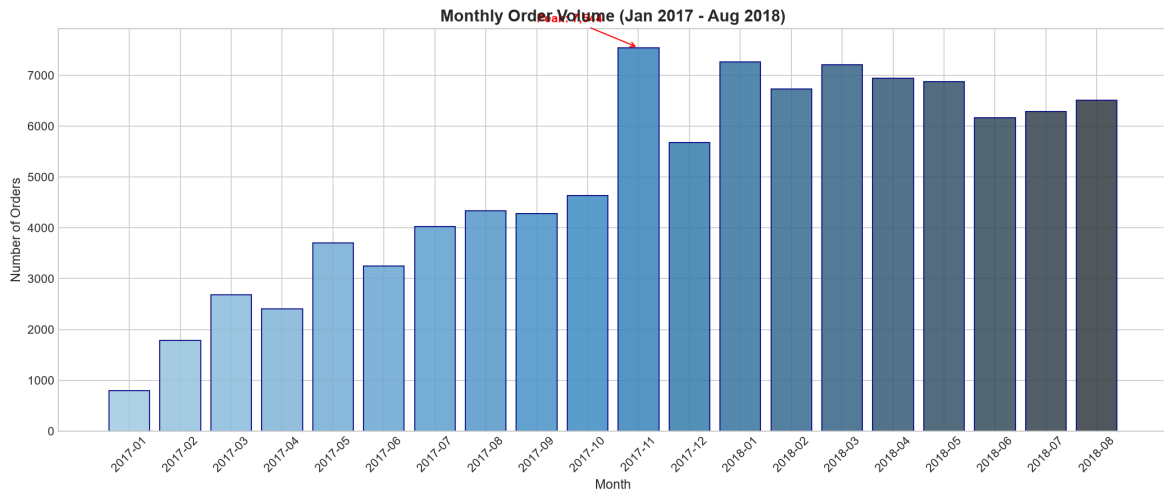


Fig C.1 Monthly order volume - growth and seasonal spike.

C.2 Distance and freight make the promise uneven

Sao Paulo holds ~42% of customers and ~60% of sellers. Same-state delivery averages ~7.5 days versus cross-state ~14.7 (+~7.2 days). Median seller-to-customer distance ~434 km (distance <-> delivery days $r \sim 0.39$). Freight totals ~R\$2.3M (~17% of product revenue); at item level, freight is ~21% of (price + freight).

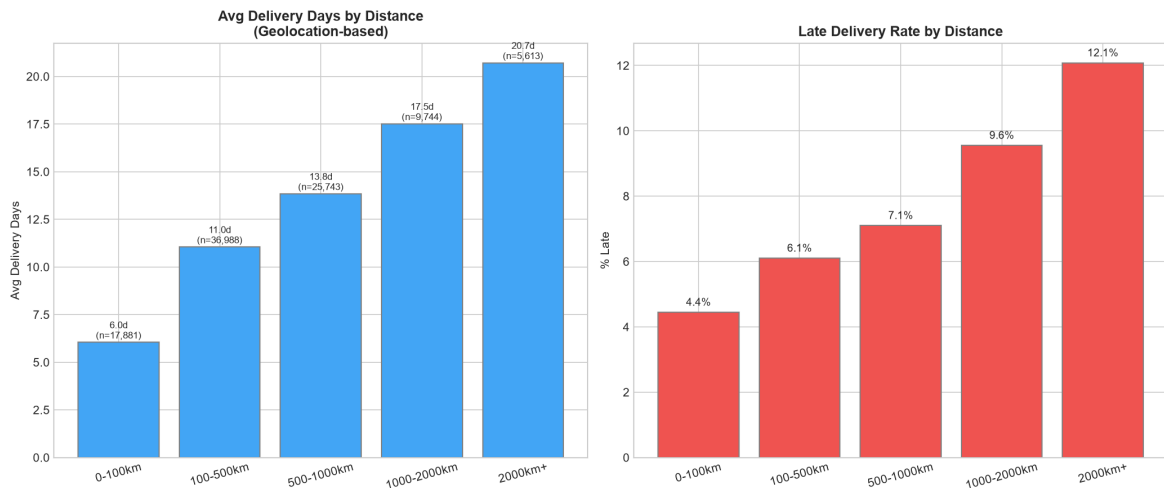


Fig C.2 Distance vs delivery - farther pairs take longer and late more often.

C.3 Late delivery destroys the shared-account score

Only 6.8% of delivered orders arrive after the ETA, but average review score falls from 4.29 (on-time/early) to 2.27 when late (-2.02 stars). Mean actual delivery ~12.1 days versus mean ETA ~23.4 - a large buffer exists, yet misses still hurt. Under aggregator logic this is reputation contagion on the shared marketplace account, not a CX footnote. Among 1-star comments (keyword scan), delivery/delay themes ~41% versus product/quality ~13%.

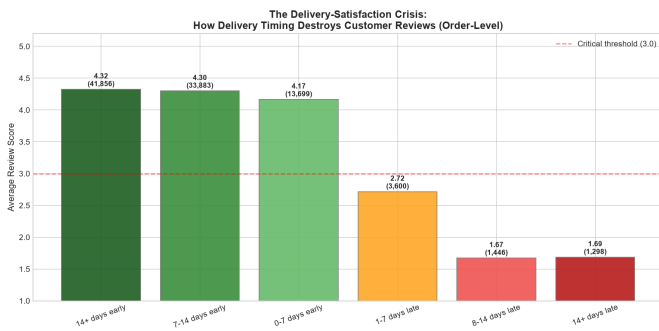


Fig C.3a Late vs on-time reviews

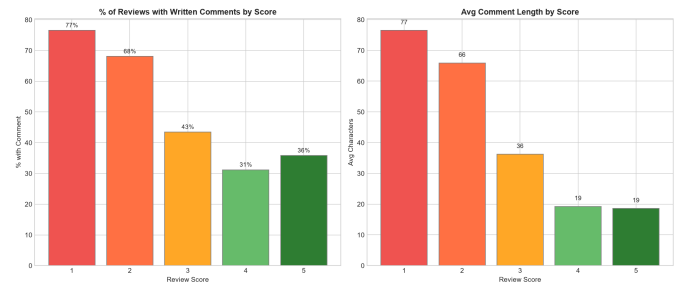


Fig C.3b 1-star text themes

C.4 Almost no loyalty + a thin seller elite

Of 96,096 unique customers, 96.9% buy once and only 3.1% return (2+ orders). Commission LTV barely compounds; one bad first delivery can erase order #2. Among 3,095 sellers, the top 10% drive ~67.5% of product revenue while the bottom 50% contribute ~3.2% - GMV dependence and long-tail reputation risk share the same Olist storefront.

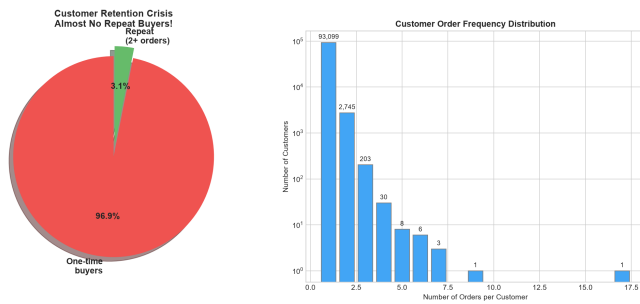


Fig C.4a One-time vs repeat buyers

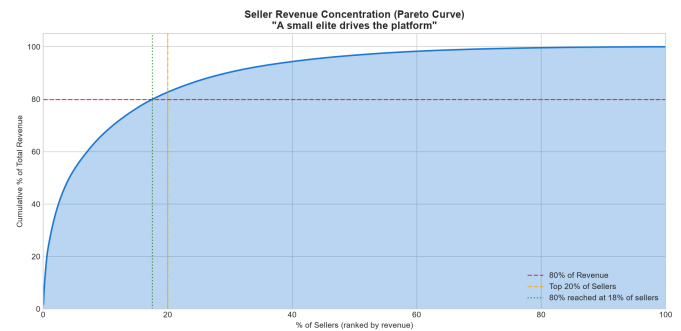


Fig C.4b Seller revenue Pareto

C.5 Priority spine (what to solve next)

#	Problem -> business stake (2016-18 model)
1	Late delivery -> review collapse shared marketplace reputation
2	Near-zero repeat purchase CAC paid once, weak LTV
3	Seller Pareto + long tail GMV fragility + contagion
4	Geo / distance SLA gap uneven national promise
5	Freight burden price perception + delay sensitivity

Bridge to HVIA solutions (next section): start from problems 1-3 - predictive delay / ETA, seller SLA risk scoring, and first-order experience protection - not from a tool list.

E. HVIA Solution Proposal

Principle from the brief: "Don't sell a tool. Find a problem worth solving." Every solution below starts from a Section C problem. Build priority follows C.5: core Solutions 1-3 first (strongest evidence, highest reputation stakes), then supporting layers that feed them, then one lighter side solution.

E.1 Solution map

#	Problem -> Solution Type HVIA domain
1	P1 Late delivery / reviews -> Predictive Late-Delivery Risk & Response Core Forecasting + Logistics Intelligence
2	P2 Near-zero repeat -> First-Order Retention Risk Flag Core Customer Intelligence
3	P3 Seller Pareto -> Seller Health Score Core Analytics & Decision + AI-Assisted Ops
4	P4 Geo/distance SLA -> Geo-aware Dynamic ETA Supporting (feeds #1) Logistics Intelligence
5	P5 Freight pressure -> Freight-Ratio Catalog Alert Side Analytics & Decision Systems
6	P6 Review-text delivery anger -> Review-Text Triage Supporting (feeds #1,#2) Customer Intelligence + AI Ops
7	P7 Growth/seasonality -> Seasonal Demand Signal Supporting (feeds #1) Forecasting & Predictive

E.2 Core 1 - Predictive Late-Delivery Risk & Response (P1)

Problem: only 6.8% of delivered orders are late vs ETA, yet reviews fall 4.29 -> 2.27 (-2.02 stars). Under the shared storefront this is reputation contagion on one Olist marketplace account - not an isolated CX issue.

What it does: at ship-prep time, score each order for probability of missing the *promised* ETA (catch cases where the generous buffer still will not be enough).

- Inputs: seller-customer distance (r~0.39 with days); same-state vs cross-state; weight/category (weight<->freight r~0.61); seller historical on-time (from Seller Health); current volume load (Seasonal Demand). Carrier field: verify in data; else proxy via freight-per-km or drop.
- Actions: Low = standard flow. Medium = proactive customer notice with a revised realistic window before the ETA is missed. High (+ poor Seller Health) = escalate to seller to expedite; flag for priority support if delay still occurs.
- KPI: shrink the on-time vs late review gap (baseline -2.02) for medium-risk notified vs control; secondary: % of high-risk orders still late after escalation (should fall).

Evidence: charts 04, 05, 06, 15.

E.3 Core 2 - First-Order Retention Risk Flag (P2)

Problem: 96.9% of customers buy once; only 3.1% return. Growth is acquisition-led; commission LTV barely compounds.

What it does: when a first order completes (delivered + reviewed, or delivered + no review after N days), score repeat-buy probability from that single experience - turning blanket "come back" campaigns into a targeted list worth a proactive touch.

- Inputs: late/high-risk flag from Solution 1; fulfilling seller Health Score; order value/category; review score if given.
- Action: high-value + high-churn-risk customers get a lightweight retention action (support check-in or small proactive gesture) - cost proportional to likely worth.
- KPI: repeat-purchase rate in the flagged high-risk/high-value cohort vs 3.1% baseline.

Evidence: chart 08.

E.4 Core 3 - Seller Health Score (P3)

Problem: top 10% of sellers -> ~67.5% product revenue; bottom 50% -> ~3.2%. Under a shared storefront, low GMV does not mean low reputation risk.

What it does: composite per-seller score (late rate, avg review, GMV contribution, growth trend). Tiers e.g. Protect & Grow / Monitor / At-Risk - any size seller whose delivery/review pattern threatens the shared account.

- Action: tier feeds Solution 1 as a delay-model feature; account/support attention prioritizes At-Risk sellers instead of spreading evenly across 3,095 merchants.
- KPI: earlier detection of reputation-risk sellers; falling share of GMV from At-Risk sellers over time as intervention works.

Evidence: chart 10. Build note: ship #3 first or alongside #1 (#1 needs the score).

E.5 Supporting components (not standalone pitches)

Scoped as accuracy layers inside Solutions 1-3 - not separate products to sell first.

E.5.1 Geo-aware Dynamic ETA (P4 -> feeds Solution 1)

Replace a flat, buffer-heavy national ETA with a per-route estimate from historical same-state vs cross-state and distance-bucket times (distance<->days $r \sim 0.39$). This is how Solution 1 knows what a realistic promise looks like for a seller-customer pair. Evidence: charts 05, 11, 12, 16.

E.5.2 Review-Text Triage (P6 -> feeds Solutions 1 & 2)

Among 1-star comments, delivery/delay themes (~41%) beat product/quality (~13%), yet median review response is ~40 hours. A lightweight classifier routes new delivery-related negatives to a priority support queue and feeds which features Solution 1 should weight. Evidence: charts 07, 18.

E.5.3 Seasonal Demand Signal (P7 -> feeds Solution 1)

Volume is highly seasonal (e.g. Nov 2017 peak 7,544 orders). During peaks, baseline late risk rises independent of one seller or route. Feed Solution 1 as a risk multiplier in forecasted high-volume windows. Evidence: charts 01, 02, 14.

E.6 Side - Freight-Ratio Catalog Alert (P5)

Freight ~17% of product revenue; item-level freight share of (price+freight) ~21%, some categories ~40% (e.g. DVDs/Blu-ray). At catalog add/reprice, flag items above a freight-to-price threshold so merchandising can intervene (bundle, min order, carrier negotiation) before weak conversion. Domain: Analytics & Decision Systems. Evidence: charts 12, 13, 17.

E.7 Suggested build order

Order	What to ship
1	Solutions 1-3 (core). Prefer Seller Health (#3) first or parallel with #1.
2	E.5.1 Geo ETA + E.5.3 Seasonal signal - upgrade Solution 1 accuracy.
3	E.5.2 Review-text triage - strengthens Solutions 1 and 2.
4	E.6 Freight catalog alert - light dependency on the rest.

One-line pitch: HVIA helps Olist protect the shared marketplace reputation and commission economics by predicting late risk before reviews collapse, focusing retention on damaged first orders, and ranking sellers by health - not by GMV alone.

End of preview (A-B-C-E). Next: Outreach draft (F), LinkedIn post (H). Full chart set (18) in outputs/charts/. Solution source: hvia_task1_section_e_solutions.md.